

CANOAS, Brazil

WMO: 839670

Lat: 29.946S

Lon: 51.144W

Elev: 8

StdP: 101.23

Time Zone: -3.00 (SAB)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.2	4.9	0.9	4.0	6.7	2.2	4.4	7.6	9.9	13.9	8.8	13.6	0.8	330	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.4	35.8	24.5	34.0	24.2	32.4	23.7	26.4	31.9	25.7	30.7	25.0	29.7	3.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.2	28.3	24.2	19.1	27.4	23.9	18.8	27.1	82.4	31.9	79.2	31.7	76.2	29.1	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.4	8.3	7.4	DB	0.1	38.5	1.1	2.2	-0.7	40.1	-1.3	41.4	-2.0	42.6	-2.8	44.2
			WB	-0.1	28.0	1.8	1.0	-1.4	28.8	-2.4	29.4	-3.4	29.9	-4.7	30.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.1	25.6	25.8	23.3	20.7	17.3	14.5	13.9	16.1	18.1	20.0	21.9	24.3
	DBStd	5.10	2.57	2.64	2.58	3.00	3.03	3.75	3.96	4.12	3.30	2.91	2.78	2.81
	HDD10.0	17	0	0	0	0	0	5	10	2	0	0	0	0
	HDD18.3	497	0	0	1	12	56	124	146	94	44	16	4	1
	CDD10.0	3703	485	443	413	320	227	141	130	192	243	309	357	444
	CDD18.3	1139	227	209	156	82	24	10	7	26	36	67	111	186
	CDH23.3	9384	2030	1862	1142	544	147	81	103	292	278	441	872	1593
	CDH26.7	3598	874	811	388	162	22	14	19	108	101	137	301	661
Wind	WSAvg	3.1	3.6	3.3	3.2	2.6	2.4	2.2	2.5	2.9	3.5	3.7	3.9	3.8
Precipitation	PrecAvg	1389	112	112	108	101	107	133	137	131	144	130	102	108
	PrecMax	1838	238	251	303	295	440	366	309	330	298	307	288	224
	PrecMin	712	10	23	15	0	2	7	19	12	15	20	8	30
	PrecStd	235	53	48	57	63	75	75	69	70	67	76	61	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.0	39.0	35.0	32.9	29.0	28.9	29.0	32.8	33.8	34.8	34.2	37.0
		MCWB	26.3	26.1	24.2	23.6	22.1	20.8	19.1	20.3	22.1	23.5	23.1	25.1
	2%	DB	35.1	35.2	32.5	30.8	26.9	26.0	26.5	29.9	29.1	30.0	32.1	34.2
		MCWB	24.8	25.8	23.7	22.7	19.9	19.8	18.6	19.9	21.2	22.0	22.7	23.8
	5%	DB	33.1	33.2	30.8	28.8	24.9	23.0	23.2	26.8	26.0	27.8	30.1	32.2
		MCWB	24.5	25.0	23.3	21.7	19.6	18.5	17.4	19.5	19.9	21.0	21.8	23.5
	10%	DB	31.2	31.2	29.0	26.8	22.9	20.2	20.1	23.2	23.8	25.8	28.1	30.7
		MCWB	24.3	24.3	22.5	20.8	18.6	17.3	16.4	18.0	18.7	20.1	21.0	23.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.9	26.0	24.6	23.2	21.8	20.8	21.7	23.3	24.2	25.8	26.6
		MCDB	33.3	34.6	32.3	30.3	26.5	26.9	25.9	28.9	29.2	31.1	30.3	32.0
	2%	WB	26.3	26.6	24.9	23.5	21.2	20.4	19.3	20.6	21.7	22.7	24.4	25.6
		MCDB	31.8	31.8	30.2	28.3	24.3	24.5	24.1	27.2	27.4	28.0	28.9	30.8
	5%	WB	25.6	25.7	24.2	22.6	20.4	19.2	18.3	19.7	20.6	21.8	23.2	24.7
		MCDB	30.5	30.3	28.6	26.6	23.2	22.4	22.2	25.4	25.0	26.2	27.1	29.4
	10%	WB	24.9	24.9	23.5	21.8	19.5	18.0	17.1	18.6	19.5	20.9	22.3	24.0
		MCDB	29.1	29.0	27.2	25.0	22.1	20.3	19.7	22.9	22.7	24.3	26.1	28.3
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.4	10.2	10.7	10.0	9.7	10.9	11.6	10.2	10.2	11.4	11.2
		MCDBR	13.5	13.2	13.1	13.4	13.3	13.9	15.8	17.3	14.9	14.0	14.9	14.3
		MCWBR	5.0	5.0	5.6	6.0	7.0	8.1	8.7	8.2	7.3	6.6	6.4	5.6
	5% WB	MCDBR	10.7	11.2	10.6	11.0	10.0	11.8	13.2	15.3	13.2	11.7	11.4	11.5
		MCWBR	4.7	4.7	4.9	5.4	5.7	7.0	7.9	7.8	6.8	6.0	5.4	4.9
Clear-Sky Solar Irradiance	taub		0.408	0.411	0.400	0.380	0.367	0.382	0.388	0.475	0.522	0.454	0.414	0.412
	taud		2.416	2.409	2.431	2.447	2.460	2.418	2.364	2.063	1.948	2.203	2.338	2.373
	Ebn at Noon		934	912	889	854	816	772	782	742	760	863	922	933
	Edh at Noon		125	122	113	100	89	89	97	145	180	149	134	131
All-Sky Solar Radiation	RadAvg		6.38	5.85	4.94	3.90	2.88	2.44	2.68	3.32	3.99	4.96	6.19	6.51
	RadStd		0.45	0.38	0.34	0.31	0.28	0.18	0.31	0.28	0.40	0.55	0.65	0.53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page